

Lead AI and ML Engineer · LLM Systems & Production GenAI · GCP · Azure · GPU Infra

Lead AI and ML engineer and data scientist with 10+ years of engineering experience across classical ML, deep learning, computer vision, NLP, forecasting, operations research, and LLM fine-tuning and serving. Design, deploy, scale, and maintain high-throughput, low-latency ML systems-from experimentation and evaluation to inference, monitoring, and retraining. Lead architecture, mentoring, and cross-functional delivery. Delivered **\$300K/year** LLM savings, retail fraud-prevention services, AWS security analytics on **billion-row datasets**, and distributed computer-vision training on **~1M images**.

Skills

- **LLM Training & Serving:** LLM Fine-tuning; LoRA / QLoRA; LLM Evaluation; OpenAI-Compatible Serving; Multi-Task Adapters
 - **Modelling & Analytical Methods:** Classical ML; Deep Learning; Computer Vision; Natural Language Processing (NLP); Time-Series Forecasting; Feature Engineering; Behavioral Anomaly Detection; Autoencoders; Operations Research; Vehicle Routing; Mixed-Integer Programming; Constraint Modelling; Supply-Chain Optimization
 - **Frameworks & APIs:** FastAPI; scikit-learn; XGBoost; PyTorch; TensorFlow / Keras; TensorFlow Serving; Hugging Face Transformers / PEFT; Unsloth; vLLM; Google OR-Tools
 - **GPU & Distributed Computing:** Distributed Multi-GPU Training; CUDA; GPU-Accelerated Computing; RAPIDS cuDF / CuPy
 - **Experimentation & Model Lifecycle:** MLflow; Experiment Tracking; Model Evaluation; Model Versioning / Registry; Automated Retraining; Vertex AI Pipelines; Azure ML Pipelines; Vertex AI Feature Store
 - **Cloud & Serverless Services:** GCP; Microsoft Azure; GKE; Vertex AI; Azure ML; Cloud Run; Cloud Functions; Azure Function Apps; Logic Apps
 - **Deployment & Production Reliability:** Docker; Kubernetes; Helm; Seldon Core; Istio; HPA Autoscaling; CI/CD; Cloud Build; Azure DevOps / Pipelines; ArgoCD; Google KCC; GitOps; JWT Authentication
 - **Monitoring, Testing & Reporting:** Prometheus; Grafana; Google Cloud Monitoring; Model Monitoring; Load Testing; Streamlit; Power BI
 - **Languages, Data & Storage:** Python; SQL; Apache Beam / Dataflow; Pub/Sub; BigQuery; MS SQL Server; Redis Caching; GCS; Azure Blob Storage; Artifact Registry; Pandas / NumPy; Dask; Git; Linux / Shell Scripting
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Experience

Lead Artificial Intelligence Engineer

Best Buy | Jul 2025 - Present

- Engineered a config-driven **LLM Fine-tuning Framework** with LoRA/QLoRA, Unsloth, and PEFT. Supported single/multi-GPU runs, multi-task adapters, and automated evaluation. YAML task configuration and standard PEFT artifacts enabled reusable tasks sharing one vLLM deployment.
- Deployed **OpenAI-compatible LLM Serving endpoints** with vLLM on GKE, HPA, and Istio; integrated GCS, Artifact Registry, Cloud Build, and Prometheus/Grafana. Self-hosting eliminated paid API dependency, saved **\$300K/year**, and strengthened data governance and compliance posture.
- Implemented **Retail Fraud Detection & Prevention** using Apache Beam/Dataflow streaming and GKE inference for XGBoost at **40,000+ events/sec** under a **200 ms SLA**. Improved fraud-score coverage by **~20%** during peak holiday traffic. Integrated Vertex AI feature, registry, and training services, secure endpoints, load testing, monitoring, and GitOps across payment, refund, and reship signals.
- Lead architecture and production delivery across search, personalization, fraud, and cybersecurity; partner with data scientists to turn validated models into secure, monitored services.

Lead ML Engineer - ML Engineering & Operations

Wipro | Aug 2023 - Jul 2025

- Engineered **AWS Account Digital Fingerprinting** training and inference with GPU-accelerated RAPIDS cuDF/CuPy on NVIDIA DGX (8xA100). Trained per-entity behavioral autoencoders on **billion-row AWS security datasets** spanning 20+ days, learning normal activity fingerprints and surfacing real-time anomalies. Integrated continuous retraining, MLflow model management, and Grafana.
- Scaled **Proofpoint Phishing Detection** pipelines to **2,400 emails/sec**, double the previous rate. Incorporated security-team feedback and Microsoft Sentinel integration to detect phishing missed by initial filters and accelerate quarantine response.
- Automated **Brand Infringement & Abuse-Inbox Analysis** through ML routing and OCR extraction of scammer identifiers across text, HTML, images, and attachments; routed evidence to security teams and mail partners for takedowns. Triage **doubled the prior case volume** and reduced manual review.
- Led and mentored ML/MLOps engineers; coordinated client/NVIDIA delivery and scaled the India AI hub through **300+ résumé screens** and **50+ technical interviews**.

Senior Data Scientist - Data Science / Azure ML

Accenture | May 2021 - Sep 2023

- Developed **Product Quality Prediction for BamaGruppen** using Azure ML warehouse-arrival scoring, with automated retraining, Logic Apps, Azure Pipelines, storage, and stakeholder alerts. Increased truck inspections from **130 to 160/day**, defect detection by **17%**, and customer approval by **20%**, delivering **\$2.5M annual savings**.
- Designed **Last-Mile Vehicle Routing** with OR-Tools and capacity, delivery-window, waiting-time, and store-demand constraints. Compared optimized routes with manual plans to improve resource utilization and reduce logistics costs.
- Implemented **Mellom Transport Optimization** with mixed-integer models for daily terminal-to-distribution-center transport, Azure Function Apps APIs, Logic Apps, RPA, CI/CD, and Slack alerts. Replaced manual Excel planning with optimized, automated, scalable plans and rapid adaptation to changing operational constraints.
- Modelled **Norsk Oppstrom Supply-Chain Optimization** scenarios across vendors, warehouses, hubs, and customers, comparing cost, freshness, and carbon footprint. Integrated simulation results with Power BI and identified the appropriate number and locations of regional hubs.
- Simulated strategic distribution hubs in the **Hub Analytics POC** to identify beneficiary customers/vendors and assess warehouse-volume reduction.
- Led cross-functional delivery, mentored data scientists, and presented findings and recommendations to Norwegian stakeholders.

Data Scientist

GD Research Centre Pvt Ltd | May 2020 - May 2021

- Developed domain-aware **GlobalData Forecasting** and scenario analysis across **15+ industry verticals**, informing strategic market-intelligence reports and resource-allocation decisions.

Deep Learning Engineer

Tata Consultancy Services | Jul 2019 - Mar 2020

- Trained **Computer Vision — Semiconductor Defect Classifiers** with distributed multi-GPU DenseNet on **~1 million images in ~3 hours**, achieving **90% accuracy** and **70% precision/recall on rare classes**. Led inference/data-platform development, deployed Docker/TensorFlow Serving at Amkor in South Korea, and trained the client team to operate the system.

Senior Software Engineer - ML

Infosys | May 2016 - Jul 2019

- Implemented **Opportunity Win/Lose Prediction** with logistic-regression scoring, SSIS data pipelines, and Power BI/SSAS dashboards for data-driven sales prioritization.
- Delivered tested **Automated ID-Card Generation** workflows for new insurance members, from impact analysis and design through implementation, support, and team knowledge sharing.

Open-Source Project

vLLM Model Chooser

Developed an interactive GPU-to-model matchmaker with VRAM, quantization, and KV-cache fit filters. [Live project](#).

Certifications

- Google Cloud Certified: Generative AI Leader | Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure Data Scientist Associate | Microsoft Certified: Power BI Data Analyst Associate
- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional

Education

B.E. Electrical & Electronics

Andhra University, Visakhapatnam | 2012 - 2016 · 80%